

EXPERIMENT 9 – STUDENT OBSERVATION

1. Write down the key features of configuring Wireless Router and DHCP Server.

Wireless Router: A wireless router provides wireless network connectivity by configuring a network name (SSID), such as **HomeNetwork**. It can provide DHCP service and use a configured DNS server address.

DHCP Server: A DHCP server automatically assigns IP addresses, subnet masks, default gateways, and DNS server information to network devices. This reduces the need for manual IP configuration.

2. What is the significance of DHCP Server in internetworking?

DHCP automatically assigns network configuration to devices, reduces manual configuration, helps prevent IP address conflicts, and makes network administration easier. Devices can obtain the required settings automatically when joining the network.

3. Design and configure an internetwork in your lab using switch, router and Ethernet cables.

The internetwork can be designed by connecting **PC0** → **Switch** → **Router** → **Switch** → **PC1** using copper straight-through Ethernet cables.

Device / Interface	IP Address	Subnet Mask	Default Gateway
Router Fa0/0	192.168.10.1	255.255.255.0	—
PC0	192.168.10.2	255.255.255.0	192.168.10.1
Router Fa0/1	192.168.20.1	255.255.255.0	—
PC1	192.168.20.2	255.255.255.0	192.168.20.1

Connectivity can be verified using the **ping** command or by sending a Simple PDU from PC0 to PC1.

Note: The uploaded lab-manual text labels these questions under Practical-10. This PDF is titled Experiment 9 as requested.